

Mathematics in Astronomy Olympiads II

Numbers, Powers, and Logarithms

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1 Introduction

Astronomy deals with numbers of extraordinary range. The mass of the Sun is about 2×10^{30} kg, while the mass of a hydrogen atom is about 1.7×10^{-27} kg, a span of nearly sixty orders of magnitude. Writing, comparing, and calculating with such numbers requires tools that go beyond ordinary arithmetic: scientific notation, the rules of exponents, and logarithms.

These tools are not merely conveniences. Some of the most important quantities in astronomy, such as the magnitude scale used to compare the brightness of stars, are *defined* through logarithms. This handout builds these ideas from the ground up. We begin with scientific notation and orders of magnitude, then develop the rules of exponents and logarithms, and finally applies them to solve problems appearing in Astronomy and Astrophysics Olympiads.

NB: As in Part 1, this handout is designed primarily for Bangladeshi middle school students, and the presentation follows the style and structure of NCTB textbooks. New ideas are introduced gradually, with practice problems to build familiarity and intuition.

2 Scientific Notation

Consider the average distance between the Earth and the Sun, written out in full:

149,600,000,000 m

Or the mass of an electron:

0.000 000 000 000 000 000 000 000 000 911 kg

Numbers written this way are difficult to read, easy to miscopy, and nearly impossible to calculate with.

Scientific notation solves this problem by expressing every number as a product of two parts: a decimal number between 1 and 10, and a power of ten. In general, we write

$$N = a \times 10^n, \quad 1 \leq a < 10,$$

where a is called the **coefficient** (or mantissa) and n is an integer called the **exponent**. The two numbers above then become

$$1.496 \times 10^{11} \text{ m} \quad \text{and} \quad 9.11 \times 10^{-31} \text{ kg}.$$

The exponent has a simple meaning: it tells us how many places the decimal point must move to recover the ordinary form of the number. A positive exponent moves the decimal point to the right (a large number), while a negative exponent moves it to the left (a small number).

To convert a number *into* scientific notation, we reverse the process: move the decimal point until exactly one

nonzero digit remains to its left, and count how many places it moved. If the decimal point moved to the left, the exponent is positive; if it moved to the right, the exponent is negative. For example,

$$299,792,458 = 2.99792458 \times 10^8 \quad (\text{decimal moved 8 places left}),$$

$$0.00052 = 5.2 \times 10^{-4} \quad (\text{decimal moved 4 places right}).$$

Problem

Express the following quantities in scientific notation:

- (a) The radius of the Sun: 696,000,000 m
- (b) The wavelength of green light: 0.00000055 m
- (c) One light-year: 9,461,000,000,000,000 m

Solution

(a) Moving the decimal point 8 places to the left leaves 6.96:

$$696,000,000 \text{ m} = 6.96 \times 10^8 \text{ m}$$

(b) Moving the decimal point 7 places to the right leaves 5.5:

$$0.00000055 \text{ m} = 5.5 \times 10^{-7} \text{ m}$$

(c) Moving the decimal point 15 places to the left leaves 9.461:

$$9,461,000,000,000,000 \text{ m} = 9.461 \times 10^{15} \text{ m}$$

Notice that these are the solar radius, the wavelength of visible light, and the light-year from Table 2 of Part 1 — quantities you will meet again and again in astronomy problems.

A useful habit: whenever a problem hands you a quantity in ordinary form, convert it to scientific notation *before* doing anything else. Every calculation that follows becomes easier, and the chance of losing a zero disappears.

3 Significant Figures

Significant figures are the digits in a number that carry meaningful information about its precision. They include all certain digits plus one uncertain or estimated digit. The number of significant figures indicates the precision of a measurement or calculation.

The following rules determine which digits in a number are significant:

1. **All nonzero digits are significant.**
742 has 3 significant figures; 2.361 has 4.
2. **Zeros between nonzero digits are significant.**
5007 has 4 significant figures; 3.02 has 3.
3. **Leading zeros are never significant.** They only locate the decimal point.
0.0052 has 2 significant figures (5 and 2 only).
4. **Trailing zeros after a decimal point are significant.**
2.500 has 4 significant figures — writing them signals that the measurement is precise to that place.
5. **Trailing zeros in a whole number without a decimal point are ambiguous.**
1500 could have 2, 3, or 4 significant figures — there is no way to tell. Scientific notation removes this ambiguity completely: 1.5×10^3 (2 s.f.), 1.50×10^3 (3 s.f.), or 1.500×10^3 (4 s.f.). This is yet another reason to prefer scientific notation.

When calculating, the result cannot be more precise than the data it came from:

- **Multiplication and division:** the result keeps as many significant figures as the *least precise* input.
 $2.34 \times 1.2 = 2.808 \approx 2.8$ (1.2 has only 2 s.f., so the answer keeps 2).
- **Addition and subtraction:** the result keeps as many *decimal places* as the input with the fewest decimal places.
 $12.52 + 1.7 = 14.22 \approx 14.2$ (1.7 is known only to one decimal place).

A full treatment of precision and uncertainty belongs to data analysis, which we will take up in a later part of this series.

Problem

The radius of a planet is measured to be $R = 3.4 \times 10^3$ km. Using $\pi = 3.14159$, calculate the volume of the planet, and express the result with the correct number of significant figures.

Solution

The volume of a sphere is

$$V = \frac{4}{3}\pi R^3$$

Substituting the given radius,

$$V = \frac{4}{3} \times 3.14159 \times (3.4 \times 10^3 \text{ km})^3$$

$$V = \frac{4}{3} \times 3.14159 \times 3.9304 \times 10^{10} \text{ km}^3$$

$$V = 1.64636... \times 10^{11} \text{ km}^3$$

Applying significant figures: The radius 3.4×10^3 km has only 2 significant figures, while π and the constant $\frac{4}{3}$ are exact numbers with unlimited precision — they do not restrict the result. Since this is a multiplication, the answer keeps as many significant figures as the least precise measurement:

$$V \approx 1.6 \times 10^{11} \text{ km}^3$$

Note that we carried the full digits through every intermediate step and rounded only at the very end. Rounding early — say, writing 3.9×10^{10} in the second line — would have risked shifting the final digit.

4 Powers and Exponents

Power is simply repeated multiplication:

$$a^n = \underbrace{a \times a \times \cdots \times a}_{n \text{ times}}$$

where a is called the **base** and n the **exponent** (or index). From this single definition, all the rules of exponents follow naturally. Let us go through them.

4.1 Rules of Exponents

1. **Product rule:** When multiplying powers with the same base, add the exponents.

$$a^m \times a^n = a^{m+n}$$

For example, $10^3 \times 10^4 = 10^7$. This is easy to see from the definition: three factors of 10 multiplied by four factors of 10 give seven factors in total.

2. **Quotient rule:** When dividing powers with the same base, subtract the exponents.

$$\frac{a^m}{a^n} = a^{m-n}$$

For example, $\frac{10^8}{10^5} = 10^3$ — five factors cancel from top and bottom, leaving three.

3. **Power of a power:** when raising a power to another power, multiply the exponents.

$$(a^m)^n = a^{mn}$$

For example, $(10^3)^2 = 10^6$.

4. **Power of a product:** An exponent distributes over multiplication.

$$(ab)^n = a^n b^n$$

For example, $(2 \times 10^4)^3 = 2^3 \times 10^{12} = 8 \times 10^{12}$.

5. **Power of a quotient:** an exponent likewise distributes over division.

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

6. **Zero exponent:** any nonzero number raised to the power zero equals one.

$$a^0 = 1$$

This is not an arbitrary convention — it follows from the quotient rule: $\frac{a^n}{a^n} = a^{n-n} = a^0$, and any number divided by itself is 1.

Note carefully that these rules apply only when the *bases are the same* (rules 1–3) or the *exponents are the same* (rules 4–5). An expression like $10^3 \times 2^4$ cannot be simplified by adding exponents — the bases differ, so each part must be evaluated on its own.

Example

Simplify the following expression using the rules of exponents:

$$\frac{(10^4)^3 \times 10^{-5}}{10^2 \times 10^3}$$

Worked Solution

Numerator: applying the power-of-a-power rule and then the product rule,

$$(10^4)^3 \times 10^{-5} = 10^{12} \times 10^{-5} = 10^{12+(-5)} = 10^7$$

Denominator: Applying the product rule,

$$10^2 \times 10^3 = 10^5$$

Combining: applying the quotient rule,

$$\frac{10^7}{10^5} = 10^{7-5} = 10^2 = 100$$

4.2 Fractional and Negative Exponents

So far we have treated the exponent as a positive whole number — a count of how many times the base is multiplied by itself. But in the previous problem we already wrote 10^{-5} without comment. What does a negative exponent mean? And can an exponent be a fraction? To answer these, we insist that the rules from the previous subsection continue to hold, and see what they force upon us.

Negative Exponents

Consider the quotient rule applied to a case where the denominator has the larger exponent:

$$\frac{10^2}{10^5} = 10^{2-5} = 10^{-3}$$

But the same quotient, worked out directly, is

$$\frac{10^2}{10^5} = \frac{10 \times 10}{10 \times 10 \times 10 \times 10 \times 10} = \frac{1}{10^3}$$

Comparing the two results, we are forced to conclude that

$$a^{-n} = \frac{1}{a^n}$$

A negative exponent therefore means *reciprocal*: it tells us to divide rather than multiply. For example, $10^{-3} = \frac{1}{1000} = 0.001$, which is exactly why small numbers acquire negative exponents in scientific notation.

Fractional Exponents

Now apply the power-of-a-power rule to an exponent of $\frac{1}{2}$:

$$\left(a^{1/2}\right)^2 = a^{\frac{1}{2} \times 2} = a^1 = a$$

So $a^{1/2}$ is the number which, when squared, gives a — that is precisely the definition of the square root. Hence

$$a^{1/2} = \sqrt{a}$$

The same reasoning extends to any root. Since $(a^{1/n})^n = a$, we have

$$a^{1/n} = \sqrt[n]{a}$$

and combining this with the power-of-a-power rule gives the general form

$$a^{m/n} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m$$

Fractional exponents appear constantly in astronomy. Kepler's Third Law, for instance, relates orbital period and semi-major axis through $T \propto a^{3/2}$ — an expression that is far easier to manipulate as a fractional power than as a root sign.

5 Logarithms

We have seen that exponents answer the question: “*what do we get when we multiply a base by itself a certain number of times?*” For instance, $10^3 = 1000$. But astronomers frequently need to ask the question the other way around: “*how many times must we multiply 10 by itself to reach 1000?*” The answer to this reversed question is called a **logarithm**.

In this section we define the logarithm carefully, establish its properties, and prepare the ground for the magnitude system — the logarithmic brightness scale that lies at the heart of observational astronomy.

5.1 What Is a Logarithm?

The logarithm of a number is the exponent to which a fixed **base** must be raised to produce that number. Formally, for a base $a > 0$ (with $a \neq 1$),

$$\log_a b = c \quad \Longleftrightarrow \quad a^c = b$$

The two equations say exactly the same thing in different forms: the left is the **logarithmic form**, the right the **exponential form**. Reading the definition aloud helps fix it in mind: “ $\log_a b$ is the power to which a must

be raised to give b .”
 For example,

$$\begin{array}{lll} \log_{10} 1000 = 3 & \text{because} & 10^3 = 1000 \\ \log_2 8 = 3 & \text{because} & 2^3 = 8 \\ \log_{10} 0.01 = -2 & \text{because} & 10^{-2} = 0.01 \end{array}$$

Two special values follow immediately from the definition and are worth memorising:

$$\begin{array}{ll} \log_a 1 = 0 & (\text{since } a^0 = 1) \\ \log_a a = 1 & (\text{since } a^1 = a) \end{array}$$

There is also an important restriction hidden in the definition. Since a positive base a raised to *any* power always gives a positive result, there is no exponent that can produce zero or a negative number. Therefore

$$\log_a b \text{ is defined only for } b > 0.$$

The logarithm of zero or of a negative number does not exist (among real numbers).

5.2 Properties of Logarithms

The real power of logarithms comes from three properties that convert multiplication into addition, division into subtraction, and powers into multiplication. Each of them follows directly from the corresponding rule of exponents from Section 4 — which is no coincidence, since a logarithm is simply an exponent in disguise. Throughout, we take the base $a > 0$ (with $a \neq 1$), and all arguments positive.

The Product Rule

$$\log_a(xy) = \log_a x + \log_a y$$

Proof:

Let $\log_a x = m$ and $\log_a y = n$. In exponential form these mean

$$x = a^m, \quad y = a^n.$$

Multiplying the two,

$$xy = a^m \times a^n = a^{m+n},$$

where the last step uses the product rule of exponents. Rewriting this in logarithmic form gives

$$\log_a(xy) = m + n = \log_a x + \log_a y.$$

The Quotient Rule

$$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$$

Proof:

With the same $x = a^m$ and $y = a^n$, divide instead of multiply:

$$\frac{x}{y} = \frac{a^m}{a^n} = a^{m-n},$$

using the quotient rule of exponents. In logarithmic form,

$$\log_a\left(\frac{x}{y}\right) = m - n = \log_a x - \log_a y.$$

The Power Rule

$$\log_a(x^k) = k \log_a x$$

Proof:

Let $\log_a x = m$, so that $x = a^m$. Raising both sides to the power k ,

$$x^k = (a^m)^k = a^{mk},$$

using the power-of-a-power rule. In logarithmic form,

$$\log_a(x^k) = mk = k \log_a x. \quad \blacksquare$$

Notice how each proof follows the same three-step pattern: rewrite the logarithms as exponents, apply the matching exponent rule, then translate back. The power rule is especially important in astronomy — it is what allows the magnitude system to turn a ratio raised to a power into a simple multiplication, as we will see in Section 6.

Warning

These rules apply to the logarithm of a product, quotient, or power — *not* to sums or differences inside the logarithm. In particular,

$$\log_a(x + y) \neq \log_a x + \log_a y.$$

There is no rule that simplifies $\log_a(x + y)$; it must be left as it is. Confusing these two situations is one of the most common mistakes students make with logarithms.

Problem

Given that $\log_{10} 2 \approx 0.301$ and $\log_{10} 3 \approx 0.477$, evaluate the following without a calculator:

- (a) $\log_{10} 6$
- (b) $\log_{10} 1.5$
- (c) $\log_{10} 8$

Solution

- (a) Write $6 = 2 \times 3$ and apply the product rule:

$$\log_{10} 6 = \log_{10} 2 + \log_{10} 3 = 0.301 + 0.477 = 0.778$$

- (b) Write $1.5 = \frac{3}{2}$ and apply the quotient rule:

$$\log_{10} 1.5 = \log_{10} 3 - \log_{10} 2 = 0.477 - 0.301 = 0.176$$

- (c) Write $8 = 2^3$ and apply the power rule:

$$\log_{10} 8 = 3 \log_{10} 2 = 3 \times 0.301 = 0.903$$

5.3 Common and Natural Logarithms

Although a logarithm can be taken to any valid base, two particular bases are so widely used that they are given their own names and notation.

The Common Logarithm

The **common logarithm** uses base 10. Because base-10 logarithms are so frequent in the physical sciences, they are usually written without showing the base at all:

$$\log_{10} x \equiv \log x.$$

Throughout this handout, and in most of astronomy, “log” with no base written means $\log_{10}$. The common logarithm is especially natural when working with scientific notation, since the logarithm of a power of ten is simply its exponent:

$$\log 10^n = n.$$

This makes the common logarithm a direct measure of a number’s *order of magnitude*. For instance, a quantity with $\log x \approx 6$ is of order a million, and the magnitude system of Section 6 is built entirely on base-10 logarithms for exactly this reason.

The Natural Logarithm

The **natural logarithm** uses the base e , an irrational constant

$$e \approx 2.71828\dots$$

and is written with its own symbol:

$$\log_e x \equiv \ln x.$$

Unlike 10, the number e is not chosen for convenience but arises naturally in the mathematics of growth and change. It can be defined precisely as the following limit:

Definition: The Number e

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

That is, e is the value that the expression $\left(1 + \frac{1}{n}\right)^n$ approaches as n grows without bound.

The number e appears throughout physics whenever rates of change are involved: in radioactive decay, in the cooling of a body, and in many other processes, which is why the natural logarithm is so important. We will return to it, and to the limit above, when this series reaches calculus.

Changing the Base

Since different problems use different bases, it is useful to be able to convert between them. Any logarithm can be rewritten in terms of another base using the **change of base formula**:

$$\log_a x = \frac{\log_b x}{\log_b a}.$$

Proof:

Let $\log_a x = c$, so that $x = a^c$. Taking the logarithm to base b of both sides and applying the power rule,

$$\log_b x = \log_b(a^c) = c \log_b a.$$

Solving for c ,

$$c = \frac{\log_b x}{\log_b a},$$

which is the formula, since $c = \log_a x$.

In practice this formula lets us evaluate a logarithm of any base using only the log (base 10) and ln (base e) keys found on every calculator. In particular, the common and natural logarithms are related by

$$\ln x = \frac{\log x}{\log e} \approx \frac{\log x}{0.4343} \approx 2.303 \log x.$$

6 Application: The Magnitude Scale

We now put logarithms to work on the scale astronomers use to describe how bright a star appears: the **magnitude scale**. Our interest here is mathematical — the physics of light, flux, and luminosity is developed separately in *Stellar Radiation I*, and readers wanting the full astronomical treatment should turn to that handout. For our purposes, it is enough to know that the **flux** F of a star is a measure of how much light we receive from it.

6.1 Why a Logarithmic Scale?

The stars visible to the naked eye differ enormously in brightness, and the faintest objects detected by modern telescopes are fainter still by many powers of ten. A scale that assigned brightness values in direct proportion to flux would therefore have to stretch across an unmanageable range of numbers.

There is also a second reason. The human eye does not judge brightness by differences but by *ratios*: a star twice as bright as another looks about as much brighter as a star four times brighter looks compared to that one. Equal ratios of flux are perceived as equal steps of brightness. A scale that turns ratios into equal steps is precisely a logarithmic scale — recall from Section 5.2 that logarithms convert division into subtraction. This is why the magnitude scale, fixed in its modern form by Norman Pogson in 1854, is defined logarithmically:

$$m_1 - m_2 = -2.5 \log\left(\frac{F_1}{F_2}\right)$$

where m_1 and m_2 are the magnitudes of two objects and F_1 , F_2 their fluxes.

Two features of this equation deserve comment. First, the minus sign means that a *brighter* object has a *smaller* magnitude, a convention inherited from the ancient catalogues. Second, the factor 2.5 is not arbitrary. It was chosen so that a difference of exactly 5 magnitudes corresponds to a flux ratio of exactly 100. Setting $m_1 - m_2 = -5$ in the equation gives

$$-5 = -2.5 \log\left(\frac{F_1}{F_2}\right) \implies \log\left(\frac{F_1}{F_2}\right) = 2 \implies \frac{F_1}{F_2} = 10^2 = 100,$$

exactly as intended. A single magnitude step therefore corresponds to a flux ratio of $100^{1/5} = 10^{2/5} \approx 2.512$.

6.2 Working with the Relation

Most magnitude problems are, mathematically, one of two tasks: either evaluating a logarithm, or undoing one. Undoing it means rewriting the equation in exponential form, exactly as we learned in Section 5.1. Rearranging Pogson's relation,

$$\log\left(\frac{F_1}{F_2}\right) = \frac{m_2 - m_1}{2.5} \implies \frac{F_1}{F_2} = 10^{(m_2 - m_1)/2.5}.$$

Problem

Two stars have apparent magnitudes $m_1 = 2.0$ and $m_2 = 7.0$. How many times brighter is the first star than the second?

Solution

We are given the magnitudes and asked for the flux ratio, so we must undo the logarithm. Starting from Pogson's relation,

$$\begin{aligned} 2.0 - 7.0 &= -2.5 \log\left(\frac{F_1}{F_2}\right) \\ -5.0 &= -2.5 \log\left(\frac{F_1}{F_2}\right) \\ \log\left(\frac{F_1}{F_2}\right) &= 2 \end{aligned}$$

Rewriting in exponential form,

$$\frac{F_1}{F_2} = 10^2 = 100$$

The first star is 100 times brighter. Note that we did not need a calculator at any stage: the magnitude difference of 5 was chosen precisely to give a round factor of 100.

The next problem is a warning as much as an exercise. It is tempting to think that combining two stars means combining their magnitudes, but magnitudes are logarithms, and logarithms do not add across a sum — as we saw in the warning box of Section 5.2.

6.3 Combined Magnitude

Magnitudes are logarithms, and this has an important consequence: they cannot be added or subtracted the way ordinary quantities can. If two stars each of magnitude 3.0 shine together, the system is *not* of magnitude 6.0, nor of magnitude 1.5. The magnitude scale is logarithmic, not linear, so arithmetic performed directly on magnitudes is meaningless — recall the warning of Section 5.2, that $\log(x + y) \neq \log x + \log y$.

What *can* be added is flux, since the light arriving from two unresolved stars simply combines. The correct procedure therefore has three stages, each of which uses a tool from the previous sections:

1. Convert each magnitude into a flux (or a flux ratio) by rewriting Pogson's relation in exponential form, as in Section 5.1.
2. Add the fluxes.
3. Convert the total flux back into a magnitude by taking a logarithm once more.

In short, we step out of the logarithmic scale, do the addition where addition is allowed, and step back in. This pattern — leave the log, operate, return to the log — appears throughout astronomy, and recognising it is often the whole difficulty of a problem.

Problem

An unresolved binary star consists of two members whose apparent magnitudes are $m_1 = 2.0$ mag and $m_2 = 4.5$ mag. What is the combined apparent magnitude of this binary system? [IOAA Junior 2023, T3]

Video Solution

The complete worked solution to this problem is presented step by step in the accompanying video, where each stage of the conversion is carried out and explained.



► Watch the Solution on YouTube

Try the problem yourself before watching.

7 Further Reading

For readers who wish to explore these topics in greater depth, the following resources are recommended:

- **Magnitude System:** NDESC, *Stellar Radiation I*
- **Flux and Magnitude Systems:** Flavio Salvati, *Fundamentals of Astronomy: A Guide for Olympiads* — Chapter 8.